

PHASE 5 – PERFORMANCE TESTING PHASE

5.1 Overview

Performance testing ensures that the **Medical Inventory Management System** built on Salesforce performs efficiently under various workloads.

The goal is to verify that the system can handle multiple users, large data volumes, and concurrent operations without any delay or failure.

This phase focuses on system speed, stability, scalability, and responsiveness to confirm it meets real-world usage demands.

5.2 Objectives

- To measure the **system's response time** during high user activity.
- To verify that the system can handle **large amounts of inventory data** smoothly.
- To check the **stability** of automation processes such as Flows, Triggers, and Reports.
- To ensure **notifications, alerts, and dashboards** load efficiently under load.
- To confirm that performance remains acceptable even when multiple users access the system simultaneously.

5.3 Types of Performance Tests Conducted

1. Load Testing

- Simulate multiple users adding, editing, and viewing inventory records simultaneously.
- Check how system performance changes as user count increases.

2. Stress Testing

- Push the system beyond normal limits (e.g., mass data upload or high report generation frequency).
- Identify breaking points and recovery behavior.

3. Scalability Testing

- Increase the number of records and users gradually to ensure the system can scale without performance issues.

4. Volume Testing

- Upload thousands of medicine records and transaction entries to measure response time and report generation speed.

5. Batch and Automation Testing

- Test scheduled **Flows, Batch Apex, and Approval Processes** under heavy data loads.

6. Response Time Testing

- Measure how quickly the system responds when users open records, generate dashboards, or run reports.

5.4 Tools and Methods Used

- **Salesforce Developer Console** – to monitor query performance and debug logs.
- **Salesforce Lightning Performance Tools** – to check page load times.
- **Data Loader** – to simulate mass record imports and observe system response.
- **Apex Test Execution** – to test automation efficiency and CPU time.
- **Reports and Dashboards** – used to monitor performance metrics visually.

5.5 Key Performance Metrics

- **Average Page Load Time:** Less than 3 seconds.
- **Flow Execution Time:** Within acceptable limits under normal loads.
- **Batch Job Completion:** Successfully completes without timeout errors.
- **Concurrent User Handling:** Supports multiple users without lag.
- **Data Processing Speed:** Large data uploads (10,000+ records) processed without system crash.

5.6 Results and Observations

- The system performed efficiently under normal and slightly high loads.
- Automated alerts and flows triggered correctly without delay.
- Reports and dashboards loaded quickly even with large datasets.
- No data loss or duplication occurred during testing.
- Minor optimization required for bulk data imports (can be improved using batch processing).